

llmXive Automated Review of Your UnEmbedding Matrix is Secretly a Feature Lens for Text Embeddings

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the validity of the mathematical derivations supporting the paper’s central hypotheses.

Mathematical Validity of the “Average Token” Derivation (Section 3.2.2): The paper claims to reverse-engineer an “average” token embedding $\hat{\mathbf{v}}_h$ using the formula $\hat{\mathbf{v}}_h = \log(\hat{\mathbf{v}}_p) \setminus, \mathbf{W}_{\mathcal{U}}^{\top}$ (Eq. 4). This derivation is mathematically flawed. The starting point is the logit equation $\mathbf{v}_h \setminus, \mathbf{W}_{\mathcal{U}}^{\top} = \log(\mathbf{v}_q) + \mathbf{b}$. The authors treat $\mathbf{b}$ as a scalar bias that can be “omitted for analytical simplicity.” However, in the context of the softmax and log-likelihood, the bias term $\mathbf{b}$ is a vector where every element is identical (the log-sum-exp term). When inverting the equation to solve for $\mathbf{v}_h$, the term $\mathbf{b} \setminus, \mathbf{W}_{\mathcal{U}}^{\top}$ does not vanish; it projects the constant bias vector onto the pseudo-inverse of the unembedding matrix. Since $\mathbf{W}_{\mathcal{U}}$ is typically rectangular ($|\mathcal{V}| \times d$) and rank-deficient (or at least not full row rank), the pseudo-inverse does not simply cancel the bias. The claim that $\hat{\mathbf{v}}_h$ is accurately recovered by ignoring this term is unsupported. The resulting $\hat{\mathbf{v}}_h$ is likely a distorted representation that does not truly correspond to the “frequency-weighted average embedding” as claimed. This undermines the entire “Logit Spectroscopy into Average Token” analysis in Section 3.2.3, as the input to that analysis is mathematically suspect.

Causal Link Between Edge Spectrum and Frequent Tokens (Section 3.2.3): The paper asserts that the “edge spectrum” (singular vectors with smallest and largest singular values) is responsible for encoding frequent tokens based on the observation that filtering these dimensions causes large logit shifts ($\Delta \pi$). While the observation of high $\Delta \pi$ at the edges is reported, the leap to the conclusion that this subspace *is* the “average token” subspace is not rigorously supported. High sensitivity ($\Delta \pi$) indicates that a dimension is important for the *current* logit distribution, but it does not prove that the dimension *encodes* the average token specifically. The paper fails to demonstrate that the projection of the *true* average token (if one could be defined independently) lies primarily in the edge spectrum. The argument relies on the circular logic that the edge spectrum is the average token subspace because filtering it changes the logits of frequent tokens, which are assumed to be the average token. A more robust claim would require showing that the edge spectrum components of *random* or *semantic* tokens are negligible, or that the edge spectrum components of the *actual* average token (derived via a different, valid method) are dominant.

Accuracy of Baseline Comparison (Section 4.2): The paper claims that the comparison with BERT-whitening is unfair because whitening “requires supervision of NLI dataset” (Section 4.2, paragraph 2). This is factually incorrect regarding the cited work (Su et al., 2021). The paper “Whitening Sentence Representations for Better Semantics and Faster Retrieval” explicitly details an *unsupervised* whitening procedure that uses only the statistics of the input data (mean and covariance) without any labeled NLI data. The authors of the current paper appear to have conflated the supervised variant of whitening with the method as a whole. By comparing their zero-shot method only against a supervised version of the baseline, the paper misrepresents the baseline’s capabilities and potentially inflates the perceived advantage of their method. The claim that “whitening... requires supervision” is false and misleads the reader about the nature

of the comparison.

These issues suggest that the core mechanistic interpretation (the “average token” and “edge spectrum” hypothesis) rests on a mathematically unsound derivation and a potentially biased experimental comparison. While the empirical performance gains of the proposed method are interesting, the *explanation* provided for *why* it works is not supported by the evidence presented.

Action items.

- **[writing]** The review focuses on the accuracy of factual claims and the validity of the mathematical derivations supporting the paper’s central hypotheses. Mathematical Validity of the “Average Token” Derivation (Section 3.2.2): The paper claims to reverse-engineer an “average” token embedding $\hat{v}_h$ using the formula $\hat{v}_h = \log(\hat{v}_p) \setminus \setminus mW_{\setminus \mathcal{U}}^{\setminus \top}$ (Eq. 4). This derivation is mathematically flawed. The starting point is the logit equation $\setminus v_h \setminus \setminus mW_{\setminus \mathcal{U}}^{\setminus \top} = \log(\setminus v_q)$

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively demonstrates the Logit Lens concept but suffers from visual clutter and the inclusion of raw escape sequences in the word clouds that are not explained in the caption.

Figure 2

The figure displays distributions for three models but is critically flawed by a missing variable definition in the caption and a logarithmic y-axis that lacks intermediate grid lines for accurate reading.

Figure 3

The figure is rendered as word clouds but is rendered unintelligible by a caption with missing variable names (indicated by a backslash). Furthermore, the specific claims regarding ‘Top-6 tokens’ and ‘colored entries’ cannot be verified against the visual data due to a lack of legends or clear visual encoding.

Figure 4

Figure 4 is scientifically unusable because it fails to label which subplot corresponds to which token category (high/low/random frequency) and does not define the y-axis metric ($\Delta\pi$). Without a legend or axis definition, the data cannot be interpreted.

Action items.

- **[science]** Figure 1: The word clouds contain raw escape sequences (e.g., ‘\n\n’, ‘\n’) and punctuation artifacts (e.g., ‘80’, ‘[’, ‘]’) that are not present in the input text provided in the caption; the visualization should either clean these tokens or the caption should explain their presence.
- **[writing]** Figure 1: The word clouds are cluttered with overlapping text and inconsistent font sizes, making it difficult to distinguish the relative importance of tokens or read smaller words clearly.
- **[fatal]** Figure 2: The y-axis label is the symbol ‘ $\Delta\pi$ ’ but the caption text is missing the symbol entirely (rendered as ‘Figure 2: () distribution...’), making the figure’s subject undefined.
- **[science]** Figure 2: The y-axis uses a logarithmic scale (0.16% to 20.0%) but lacks grid lines or tick marks between the labeled decades, making it impossible to estimate intermediate values.
- **[writing]** Figure 2: The x-axis tick labels (e.g., 112, 224, 336) are non-standard and lack a clear unit or definition (e.g., ‘dimension index’ or ‘tokens’), which is not explained in the caption.
- **[fatal]** Figure 3: The caption contains critical missing information, specifically the name of the method or variable represented by the backslash symbol (\) in ‘refined by ’ and ‘\ suppresses...’. This makes the figure’s context and claims unintelligible.
- **[science]** Figure 3: The caption claims to display ‘Top-6 tokens’, but the word clouds show dozens of tokens of varying sizes without a clear legend or scale defining the top 6, making the specific claim unverifiable.
- **[science]** Figure 3: The caption states ‘colored entries indicate tokens that have literal connections’, but the word clouds use a multi-color palette for all words without a legend or visual distinction to identify which specific tokens are ‘colored entries’ versus others.
- **[fatal]** Figure 4: The y-axis label is the symbol ‘ $\Delta\pi$ ’ without defining what this metric represents (e.g., change in probability, divergence). The caption refers to ‘ $\Delta\pi$ distribution’ but does not define the variable, making the scientific content unintelligible.
- **[fatal]** Figure 4: The figure contains three subplots but lacks a legend or any labels to distinguish the three conditions mentioned in the caption (‘high-frequency’, ‘low-frequency’, and ‘randomly sampled tokens’). It is impossible to determine which subplot corresponds to which token category.

- **[science]** Figure 4: The y-axis uses a logarithmic scale (0.16% to 20.0%) but lacks grid lines or tick marks at intermediate powers of 10, making it difficult to accurately compare the magnitude of values across the different subplots.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology from mechanistic interpretability and linear algebra without providing sufficient scaffolding for a general audience. While the core concepts are sound, the density of jargon creates a barrier to entry.

Specifically, the terms “Logit Lens” and “Logit Spectroscopy” are introduced in the Introduction (Section 1) with a footnote directing readers to Section 2 for details. This is insufficient for a general paper; these tools should be briefly defined in plain language at their first mention (e.g., “a technique to project hidden states back to the vocabulary space”). Similarly, “anisotropic” is used repeatedly (Sections 1, 3, 5) to describe the geometry of embeddings. While the paper attempts to explain this as “confined to a narrow cone,” the term itself is jargon that should be either defined immediately or replaced with the descriptive phrase throughout the text to ensure clarity.

The concept of the “edge spectrum” is central to the paper’s contribution but is introduced with a dense mathematical definition in the Introduction. The phrase “actively writing these frequent tokens into embedding space” (Abstract, Section 1) uses the specific “writing” metaphor common in interpretability literature but is opaque to non-specialists. This should be rephrased to “biasing the representation toward” or “encoding” for broader accessibility.

Finally, the acronym “MTEB” appears in Section 5 without prior expansion, despite the full name “Massive Text Embedding Benchmark” being mentioned in the abstract. All acronyms must be defined at first use. The paper would benefit from a “Notation” or “Glossary” section, or simply more consistent plain-language definitions for these key terms to ensure the findings are accessible to the broader NLP community.

Action items.

- **[writing]** The manuscript relies heavily on specialized terminology from mechanistic interpretability and linear algebra without providing sufficient scaffolding for a general audience. While the core concepts are sound, the density of jargon creates a barrier to entry. Specifically, the terms “Logit Lens” and “Logit Spectroscopy” are introduced in the Introduction (Section 1) with a footnote directing readers to Section 2 for details. This is insufficient for a general paper; these tools should be briefly

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical consistency of the paper is compromised by a critical error in the mathematical proof regarding dimensionality reduction and a gap in the derivation of the “average” token.

First, the proof in Appendix \ref{sec:proof} (lines 1040-1055) contains a fundamental mathematical flaw. The authors state: “considering that V_{τ} , $V_{\tau}^{\top}$ is identity, thus we have: $\|vz\|, \|V_{\tau}\|, \|V_{\tau}^{\top}\|_2 = 1$; $\|vz\|_2$.” This is incorrect. V_{τ} consists of a subset of columns from the orthogonal matrix V . Therefore, V_{τ} , $V_{\tau}^{\top}$ is a projection matrix onto the subspace spanned by those columns, not the identity matrix. The norm $\|vz\|, \|V_{\tau}\|, \|V_{\tau}^{\top}\|_2$ is generally strictly less than $\|vz\|_2$ unless vz lies entirely within that subspace. The claim that the transformation preserves distances *while* reducing dimensionality relies on a false premise in the provided proof.

Second, the derivation of the “average” token in Section 3.2 (lines 630-650) relies on the equation $\hat{v}_h = \log(\hat{v}_p)$, $W_{\mathcal{U}}^{\top}$. This equation assumes a linear relationship between the log-probabilities of the frequency distribution and the hidden state, derived by inverting the Softmax operation. However, the Softmax function is non-linear, and the relationship $v_q = \operatorname{Softmax}(v_h, W_{\mathcal{U}}^{\top})$ cannot be simply inverted to $v_h = \log(v_q)$, $W_{\mathcal{U}}^{\top}$ without accounting for the normalization constant (the log-sum-exp term) which acts as a global bias. While the authors mention omitting the bias term v_b for simplicity, the omission of the normalization term in the log-domain inversion is a significant logical gap that undermines the claim that $\hat{v}_h$ is a precise “reverse-engineered” representation of the average token.

Finally, the ablation study in Table 4 presents a logical tension with the proposed method. The method filters both the “Dominant” (largest singular values) and “Secondary” (smallest) subspaces. However, the ablation shows that filtering only the “Secondary” subspace (Config 4) yields results (Avg 53.19) much closer to the full method (Avg 54.57) than filtering the “Dominant” subspace (Config 3, Avg 47.53) or the “Bulk” (Config 5, Avg 47.13). The paper argues that the “edge spectrum” (both ends) is responsible for the “average” token effect, but the data suggests the “Dominant” edge might not be the primary culprit for the specific degradation being fixed, or that its removal is less critical than the “Secondary” removal. The justification for including the “Dominant” subspace in the filter is not logically derived from the ablation results provided.

Action items.

- **[fatal]** The proof in Appendix sec:proof claims $V_{\tau} * V_{\tau}^{\top}$ is the identity matrix, which is false for a truncated projection. This invalidates the step $\|z * V_{\tau} * V_{\tau}^{\top}\| = \|z\|$. The proof must be corrected to show norm preservation via orthogonality of V_{τ}

in the reduced space.

- **[science]** The derivation of the ‘average’ token ($\hat{h} = \log(\hat{p}) * W_U^+$) ignores the non-linearity of Softmax and the log-sum-exp normalization term. The linear inversion used is mathematically unsound, making the ‘reverse-engineering’ claim speculative rather than derived.
- **[science]** The ablation study shows filtering only the ‘Secondary’ subspace (Config 4) performs nearly as well as the full method, while filtering ‘Dominant’ (Config 3) hurts performance. The logic for including the ‘Dominant’ subspace in the filter is not justified by these results.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that overreach the provided evidence, particularly regarding the mathematical properties of the proposed transformation and the extent of performance preservation.

First, the claim in Section 4.2 and the Appendix that the proposed transformation is “distance-preserving” and that reducing dimensions “causes no theoretical difference in similarity measurement” is mathematically unsound. The proof in Appendix B (Section “Equivalence Transformation Proof”) contains a critical error: it asserts that $\mathbf{M}_V \mathbf{V}_\tau \mathbf{M}_V^T$ is the identity matrix. Since $\mathbf{M}_V \mathbf{V}_\tau$ consists of a subset of columns from the orthogonal matrix $\mathbf{M}_V$, $\mathbf{M}_V \mathbf{V}_\tau \mathbf{M}_V^T$ is a projection matrix, not the identity. Consequently, $\|\mathbf{v}_z\|_{\mathbf{M}_V \mathbf{V}_\tau \mathbf{M}_V^T}^2 \neq \|\mathbf{v}_z\|_2^2$ for vectors $\mathbf{v}_z$ with components outside the subspace. The authors overreach by claiming theoretical equivalence where a projection error is inevitable. This invalidates the claim that dimensionality reduction is “free” in terms of distance preservation.

Second, the abstract and introduction state that the method allows for dimensionality reduction “while fully preserving the refined embedding quality.” The experimental results in Table 1 contradict the word “fully.” For instance, with Qwen2.5-0.5B and ECHO, the average score drops from 52.55 ($\tau=2$) to 49.43 ($\tau=8$), a significant degradation. While the method improves over the baseline, claiming “full preservation” when performance drops by ~6% at higher compression ratios is an overstatement of the empirical findings.

Finally, the mechanistic interpretation relies heavily on the analysis of a reverse-engineered “average” token. While the Logit Spectroscopy on this synthetic token shows high sensitivity in the edge spectrum, the paper over-extrapolates this to claim that this subspace is the primary driver of anisotropy in *actual* text embeddings without providing a direct causal link or ablation showing that removing this subspace from real embeddings is the *only* reason for the

improvement, rather than a general regularization effect.

Action items.

- **[science]** The claim that the transformation is ‘distance-preserving’ (Section 4.2, Eq. 1) is mathematically incorrect for dimensionality reduction. The proof in Appendix B incorrectly assumes $V_{\tau} * V_{\tau}^T$ is the identity matrix; it is a projection matrix. Consequently, the claim that reducing dimensions to $1/\tau$ ‘causes no theoretical difference in similarity measurement’ is an overreach not supported by the provided proof.
- **[writing]** The abstract and introduction claim the method enables ‘inherent dimensionality reduction... while fully preserving the refined embedding quality.’ The results in Table 1 show performance drops (e.g., Qwen ECHO Avg drops from 52.55 to 49.43 at $\tau=8$). The claim of ‘fully preserving’ quality is an overstatement given the observed trade-offs in the data.
- **[science]** The paper asserts that the ‘edge spectrum’ is ‘primarily responsible for encoding high-frequency tokens’ based on Logit Spectroscopy of a reverse-engineered ‘average’ token. However, the paper does not provide evidence that this specific subspace is the *sole* or *dominant* cause of anisotropy in actual text embeddings versus the synthetic average token, potentially over-extrapolating from a proxy analysis.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This review focuses exclusively on safety, ethics, and potential dual-use risks associated with the proposed “EmbedFilter” (EmbFilter) method.

The paper presents a novel technique to improve LLM text embeddings by filtering out subspaces associated with high-frequency tokens. While the primary goal is to enhance semantic representation and reduce dimensionality, the mechanism itself introduces specific safety and ethical considerations that require clarification.

First, the **Conflict of Interest (COI)** disclosure is insufficient. The Acknowledgments section (lines 1045-1050) reveals that the work is supported by Lenovo Group, and two authors (Heng Cui, Cong Li) are affiliated with Lenovo. Given that the method involves modifying how models represent text, which could have significant implications for enterprise search and content moderation, a formal COI statement detailing the nature of the funding and any potential commercial interests is necessary to maintain transparency.

Second, the **dual-use potential** of the method warrants discussion. The core mechanism involves identifying and suppressing “high-frequency” tokens to reduce anisotropy. While the authors frame this as removing “uninformative” tokens (e.g., “the”, “a”), the definition of “frequent” is relative to the training distribution. A malicious actor could potentially adapt this technique to systematically suppress specific keywords, political viewpoints, or safety-critical

terms by manipulating the frequency distribution or the “average token” definition. This could be used for covert censorship or to bias retrieval systems against certain topics. The authors should address this risk in the discussion or conclusion, perhaps by suggesting safeguards or limitations on the method’s application.

Third, the **methodological reliance on proxy data** raises ethical questions regarding reproducibility and unintended consequences. The authors approximate the true training distribution using the RedPajama dataset (Section 3.2.1, lines 430-435) because the actual pretraining data is proprietary. If the proxy distribution differs significantly from the actual training data, the “edge spectrum” identified may not accurately represent the model’s true biases or frequent tokens. This could lead to the unintended suppression of legitimate semantic content or the failure to remove actual noise, potentially degrading the model’s performance or introducing new biases. The authors should explicitly discuss the limitations of using a proxy dataset for this reverse-engineering process and the potential ethical implications of deploying a method based on such approximations.

Finally, the paper does not mention **data privacy** concerns. While the method operates on model weights and public benchmarks, the underlying logic of identifying “average” tokens relies on assumptions about the training data. If the method were adapted to analyze private or sensitive datasets, the reverse-engineering process could potentially leak information about the training distribution. A brief statement on the privacy implications of analyzing model internals in this manner would be prudent.

In summary, while the technical contribution is sound, the manuscript requires minor revisions to address conflicts of interest, potential dual-use risks, and the ethical implications of using proxy data for model interpretation.

Action items.

- **[writing]** The Acknowledgments section (lines 1045-1050) explicitly states the work is supported by Lenovo Group and lists authors from Lenovo. The manuscript must include a formal Conflict of Interest (COI) statement clarifying the nature of this funding and any potential influence on the research design or results, as required by most academic venues.
- **[writing]** The paper proposes a method to filter ‘edge spectrum’ subspaces to remove high-frequency tokens. While intended to improve semantic quality, this mechanism could be misused to systematically suppress specific topics, political viewpoints, or safety-critical keywords if the ‘frequent token’ definition is manipulated. The authors should add a discussion on the potential for dual-use in censorship or bias amplification and propose mitigation strategies.
- **[writing]** The methodology relies on approximating training corpus word frequencies using the RedPajama dataset (Section 3.2.1, lines 430-435) because the actual pretraining data is undisclosed. The authors should explicitly discuss the ethical implications of using a proxy dataset for reverse-engineering model internals, particularly if the proxy distribution differs significantly from the actual training data, potentially leading to unintended filtering of legitimate semantic content.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents compelling empirical evidence that filtering the edge spectrum of the unembedding matrix improves text embedding quality. The use of Logit Lens and Logit Spectroscopy provides a strong mechanistic narrative, and the results across multiple LLM backbones (Qwen, Llama, Mistral) are consistent and robust. The ablation studies effectively demonstrate that the gains are not merely due to dimensionality reduction.

However, the scientific evidence is undermined by a critical mathematical error in the proof of the distance-preserving property (Appendix~\ref{sec:proof}). The authors claim that $V_{\tau} V_{\tau}^T$ is the identity matrix, which is only true if V_{τ} is square. Since the method explicitly reduces dimensions ($\tau > 1$), V_{τ} is rectangular, and the product is a projection matrix, not the identity. Consequently, the Euclidean distance between vectors is not preserved in the reduced space as claimed. This error invalidates the theoretical justification for the dimensionality reduction claim and must be corrected.

Additionally, the reverse-engineering of the “average” token relies on a heuristic inversion of the unembedding matrix that assumes the bias term is negligible. While the empirical results are convincing, the theoretical robustness of this construction across different model architectures and vocabulary sizes requires stronger justification. Finally, while the ablation studies show that filtering the “bulk” spectrum is detrimental, the paper lacks a deep theoretical explanation for why the edge spectrum specifically encodes high-frequency tokens, leaving the mechanism somewhat empirical. Addressing these points will significantly strengthen the scientific validity of the claims.

Action items.

- **[science]** The proof in Appendix~\ref{sec:proof} claims $V_{\tau} * V_{\tau}^T$ is identity, which is false for dimensionality reduction ($\tau > 1$). This invalidates the ‘distance-preserving’ claim. Correct the proof to reflect the projection geometry.
- **[science]** The ‘average’ token reverse-engineering assumes the unembedding matrix pseudo-inverse recovers a hidden state from log-probs, ignoring bias effects. Provide stronger justification or empirical validation for this heuristic across model architectures.
- **[science]** The ablation study shows ‘Bulk’ filtering fails but lacks a theoretical explanation for why edge spectra specifically encode high-frequency tokens. Add mechanistic analysis to rule out overfitting to specific model spectral properties.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis and mathematical derivations supporting the core claims of the paper require significant revision.

First, the proof of the distance-preserving property in Appendix~\ref{sec:proof} (lines 1080-1095) is mathematically incorrect. The authors state that “considering that $V_{\text{tau}} * V_{\text{tau}}^T$ is identity,” which is false. Since V_{tau} consists of a subset of columns from an orthogonal matrix V , $V_{\text{tau}} * V_{\text{tau}}^T$ is a projection matrix onto a subspace, not the identity matrix. Consequently, the equality $\|z * \Phi^T\|_2 = \|z\|_2$ does not hold; the norm is strictly reduced unless z lies entirely within the subspace. This invalidates the claim that the transformation is distance-preserving in the Euclidean sense, which is central to the dimensionality reduction argument in Section 4.2.

Second, the experimental results in Table~\ref{tab:main_results} and Table~\ref{tab:ablation} lack necessary statistical rigor. The paper reports mean performance across 49 datasets but provides no measure of variance (e.g., standard deviation) or statistical significance testing (e.g., paired t-tests or bootstrap confidence intervals). Given the magnitude of the reported improvements (up to 14.1%), it is essential to demonstrate that these gains are statistically significant and not artifacts of the specific dataset split or random initialization. Without this, the robustness of the method across different tasks cannot be verified.

Finally, the derivation of the “average” token in Section 3.2.2 relies on the approximation $\log(\text{frequency}) \approx \text{logit}$. This ignores the softmax normalization constant and the bias term v_b , which the authors admit to omitting. While the bias might be constant, the normalization term varies with the vocabulary distribution. A more rigorous statistical justification or sensitivity analysis regarding this approximation is needed to ensure the identified “edge spectrum” is not an artifact of the simplified logit estimation.

Action items.

- **[fatal]** The proof in Appendix~\ref{sec:proof} contains a fatal algebraic error. The authors claim $V_{\text{tau}} * V_{\text{tau}}^T$ is the identity matrix, which is false for a truncated projection matrix (it is a projection, not identity). Consequently, the claim that $\|z * \Phi^T\|_2 = \|z\|_2$ is incorrect; the transformation reduces the norm. The dimensionality reduction claim relies on this flawed identity.
- **[science]** The ablation study in Table~\ref{tab:ablation} lacks statistical significance testing. With 49 datasets, reporting single-point averages without standard deviations, confidence intervals, or paired t-tests makes it impossible to determine if the observed gains (e.g., +9.0%) are robust or due to random variance.
- **[science]** The reverse-engineering of the “average” token in Section~\ref{sec:interpret} uses $\log(\text{frequency})$ as a proxy for logits without accounting for the temperature scaling or the

bias term (vb) omitted in Equation 5. This approximation may introduce systematic bias in the identification of the edge spectrum subspace.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling technical contribution, but the writing quality requires minor revisions to meet publication standards. The primary issues involve subject-verb agreement errors, article misuse, and minor typographical errors that interrupt the flow of reading.

Specifically, in the Abstract and Section 1, the authors write “By filtering out this subspace, \name\ suppress the influence,” where the singular subject “\name” requires the verb “suppresses.” This same error appears in Section 1 regarding the “mechanism interpretation,” which incorrectly uses the article “an” instead of “a.” In Section 3.1, the phrase “preliminaries analyses” is grammatically incorrect and should be “preliminary analyses.” Furthermore, a sentence in Section 1 ends abruptly with “identification” without a period, creating a fragment.

In the results discussion (Section 5.2), the sentence “prompt-engineering methods exhibits performance fluctuations” contains a plural-singular mismatch; “methods” should pair with “exhibit.” A simple typo, “acorss,” appears in Section 5.3 and should be corrected to “across.” Finally, in the Appendix proof, the concluding sentence begins with a lowercase “this,” which should be capitalized.

Addressing these grammatical inconsistencies and typos will significantly improve the professionalism and readability of the paper. The logical flow and structure of the arguments are otherwise sound.

Action items.

- **[writing]** In the Abstract, the phrase ‘By filtering out this subspace, \name\ suppress the influence’ contains a subject-verb agreement error. Since \name\ is singular, it should be ‘suppresses’. This error appears in the Abstract and Section 1.
- **[writing]** In Section 1 (Introduction), the sentence ‘We provide an mechanism interpretation’ uses the incorrect article ‘an’ before a consonant sound. It should be ‘a mechanism interpretation’.
- **[writing]** In Section 1, the sentence ending ‘...providing a detailed account of its identification’ is a fragment lacking a period. Additionally, the phrase ‘preliminaries analyses’ in Section 3.1 should be ‘preliminary analyses’.
- **[writing]** In Section 5.2, the phrase ‘...prompt-engineering methods exhibits performance fluctuations’ has a subject-verb agreement error. ‘Methods’ is plural, so it should be ‘exhibit’.
- **[writing]** In Section 5.3, the word ‘acorss’ is a typo and should be corrected to ‘across’.

- **[writing]** In the Appendix (Section A.2), the proof concludes with ‘this completes the proof...’, which should be capitalized as ‘This completes the proof...’ to start a new sentence.